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About this document

This is the **submission version** of the final report, written to the per-chapter word limits. It is derived from `final/final_report.md`, which is the project's full research record and is maintained continuously as development proceeds. Nothing here contradicts that document; where this version states a result, the record holds the derivation, the intermediate measurements and the failures that preceded it. Appendices carry evidence that does not fit the caps.

Word counts are checked by `final/wordcount.py`, which reports the counted prose separately from the raw total so that the reliance on the brief's exclusion of tables and figures is visible rather than assumed.

Abstract

Post-bout video review in fencing is manual across the whole sport. This project develops an AI-assisted web application that chains three pre-trained models, YOLOv8 detection with ByteTrack, MediaPipe Pose and a large language model, into a pipeline that turns an uploaded epee bout into a distance series, proposed touches, per-fencer measurements and a written summary, which a user then confirms or corrects through a browser interface. Touch detection reaches F1 0.85 across three clips against hand-labelled ground truth, and reading the scoring machine's lamps identifies whether one named fencer was involved in all twenty-seven labelled touches. The project's most transferable outcome is methodological: five separate metrics passed every test written for them while measuring something other than what they claimed, and each was exposed by measurement against an external constraint rather than by further reasoning.

1. Introduction

Context

Video review is routine in sports performance analysis, but in fencing the review itself has remained manual across the whole sport rather than only at its amateur end. Coaches at elite, FIE and Olympic level still scrub footage by hand and tag actions themselves; no automated pipeline is in widespread use anywhere in the sport. What differs by level is the resources brought to the same manual task, not the task. At club level the consequence is that recorded bouts often sit unreviewed altogether, or are reviewed by watching whole clips repeatedly for qualitative impressions, and even dedicated fencing analysis tools require the user to label every action by hand.

The addressable need is therefore broader than accessibility for amateurs, although amateur accessibility is this project's primary user focus: a club fencer has no analyst and no budget, so they are the user for whom automation makes the difference between analysis happening and not happening.

Motivation

Several tactically valuable quantities are measurable by hand only live, at the cost of a coach's full attention, and then only as coarse discrete events. Inter-fencer distance over time, engagement distance and its variance, and the frequency of advances, retreats and lunges cannot be recorded continuously by eye at all. This layer of quantitative insight is consequently almost never captured, and review remains limited to subjective impressions.

Problem statement and aims

The problem is to design and build a working system that ingests an uploaded epee bout video; uses pre-trained models to extract fencer positions, poses, derived movement metrics and likely event segments; presents that information through an annotation interface allowing confirmation, correction and labelling; and produces aggregated outputs that are useful to fencers and coaches.

A central design principle is that the system is AI-assisted, not automatic. The unreliability of computer vision on real fencing footage is treated as a constraint the workflow accommodates rather than a problem to be solved before the system can be useful.

What this adds that existing tools do not

Stated plainly, because the obvious reading of a project like this is that it automates work other systems already do, and that is not the claim.

Existing fencing tools are **manual loggers**. The user watches the bout and tags what they saw; the analytics that follow are summaries of those tags. What such a system knows is exactly what was typed into it. This project's measurement layer knows things nobody entered: the distance between the fencers at every frame of the recording, and every quantity derived from it.

That difference is not one of convenience. **Continuous inter-fencer distance is not merely absent from manual tools; it is impractical to produce by hand at all**, because recording it

for one bout means pausing and measuring across thousands of frames. A coach can log who scored and roughly where they were standing. No coach can log the distance thirty times a second, and so no manual system, however well designed, can report how a fencer's working distance changed under pressure or where on the strip their touches were scored from.

The event layer is the opposite case, and the report does not overclaim it. Deciding that a touch occurred, and to whom, is a judgement a coach makes accurately and instantly, and the system's automated attempts at it are worse. Those are proposals precisely because the human is better at them. **The contribution is therefore a division of labour rather than a replacement:** measurement that cannot be done by hand, paired with judgement that should not be taken from the user.

Contributions

A working end-to-end system. A bout is uploaded through a browser, processed by a background job runner and reviewed through an annotation interface, with no terminal involved. Three pre-trained models are chained rather than demonstrated separately: detections bound the region pose works in, pose landmarks supply the ground reference the distance series is measured from, that series drives touch proposal, confirmed touches scope every aggregate, and those aggregates are what a language model interprets.

A geometric touch detector evaluated against hand-labelled ground truth. Twenty-seven touches were labelled across four independently sourced bouts, with clips held back from tuning so generalisation could be tested rather than assumed. An audio detector was built first, shown to fail on held-back footage, and replaced.

Two capabilities the detector could not supply. Attribution of a touch to a fencer, read from the scoring machine's lamps, and lunge proposals calibrated per bout from lunges the user has already confirmed. Both are reported with the limits measurement established.

A methodological result about measuring one's own system. Five metrics survived their own derivation and then failed a test of what they could actually detect. Chapter 5 treats that pattern as a finding, because the diagnosis in each case came from measurement against an external constraint rather than from further reasoning.

Scope

Out of scope by choice: refereeing decisions and right of way, which epee does not require; identification of individual athletes, which the system avoids by tracking anonymous slots; and multi-camera or three-dimensional reconstruction, which would resolve several measurement limitations and is treated as further work.

This project corresponds to the CM3020 template "Orchestrating AI Models to Achieve a Goal", which expects an integrated system built around at least three pre-trained models.

Structure of this report

Chapter 2 reviews the literature and positions the project against existing systems. Chapter 3 gives the design, including the argument that settles the system's role as an assistant rather than an automaton. Chapter 4 is written as a development record rather than a description of a finished

artefact, because the changes of direction carry most of the evidence. Chapter 5 reports results on real footage and, equally, the results that did not survive scrutiny. Chapter 6 judges the outcome against the aims above and states what the project failed to establish.

2. Literature Review

This chapter reviews work across four themes relevant to the application: deep learning in sports video analysis; temporal action detection in untrimmed video; pose estimation for fine-grained sports actions; and the human-in-the-loop paradigm underpinning the assisted design. It then examines existing systems in order to position the project. Where the implementation has since tested a claim from this literature against real footage, the outcome is noted, because a review that informed a build should be readable alongside what the build found.

2.1 Deep learning in sports video analysis

Rangasamy et al. (2020) contrast handcrafted-feature approaches with deep learning for human activity recognition in sport, establishing that convolutional networks dominate frame-level understanding, with recurrent and temporal models capturing motion across sequences. The value here is methodological: it confirms that deep learning is the appropriate family of techniques for extracting structure from sports footage. The review nonetheless treats analysis as a fully automatic classification problem and is necessarily broad rather than fencing-specific. It does not address how such systems should behave where reliable automation remains out of reach, which is precisely this project's situation.

2.2 Temporal action detection in untrimmed video

Vahdani and Tian (2021) survey deep-learning approaches to localising when actions occur within long untrimmed footage. This bears directly on the system, because a bout is a continuous recording in which touches are sparse and separated by footwork, preparation and pauses. Their taxonomy of supervision levels is instructive: the fully supervised methods that dominate published benchmarks depend on dense frame-level boundary annotations that are expensive to produce and that do not exist for any fencing dataset. The survey is explicit that temporal action detection remains an open research problem rather than a solved capability.

Two implications follow. Automatic event localisation should highlight probable segments for human review rather than serve as a reliable end in itself; and the absence of annotated fencing data makes a fully supervised automatic system infeasible within an undergraduate project. The realistic contribution is to reduce manual review time through assisted highlighting.

2.3 Pose estimation for fine-grained sports actions

Hong et al. (2021) introduce Video Pose Distillation and demonstrate that pose is a strong cue for fine-grained sports action recognition. Equally importantly, they show that off-the-shelf pose estimators degrade in sports footage through motion blur, occlusion and domain shift. This applies acutely to fencing, where two athletes overlap on a narrow piste and move explosively. Their conclusion that pose is informative yet unreliable supports a design in which pose-derived features inform suggestions a user confirms.

The implementation bears this out. Pose succeeded on roughly 30 to 32 per cent of frames where both fencers were tracked, a figure governed mainly by a deliberate stride of three frames, and it failed most often during precisely the clinches and rapid exchanges Hong et al. identify. The pipeline falls

back to bounding-box geometry on those frames rather than treating a missing pose as missing data. Their distillation method is, however, a model-training contribution aimed at improving accuracy on curated datasets; it does not deliver an end-user tool, and reproducing its training pipeline lies beyond this scope. This project therefore adopts pose as a pragmatic pre-trained component rather than attempting to advance pose recognition itself.

2.4 Fencing-specific computer vision

The closest domain-specific work is Mo (2022), whose “Allez Go” applies pose estimation and a temporal convolutional network, augmented with audio, to referee bouts. It reports approximately 89 to 90 per cent accuracy in classifying which fencer scored, on a custom dataset of around 4,000 international-level clips, with audio detecting blade contact. This is evidence that pose-based fencing analysis is feasible and that audio can complement visual features.

Two contrasts matter. Allez Go pursues automatic refereeing, a high-stakes binary decision, whereas this system pursues assisted profiling, a lower-stakes task that tolerates imperfect automation far better. And Allez Go trains on elite, well-filmed competition footage. It would be wrong to infer that manual review is an amateur problem: it remains manual at FIE and Olympic level too. What elite footage provides is better conditions for automation, namely fixed camera positions, consistent framing and professional lighting. Club recordings are single-camera, less standardised and noisier, weakening the case for full automation further. Allez Go therefore simultaneously demonstrates feasibility and reinforces the rationale for an assisted design, with the assisted approach mattering most where footage quality is lowest.

2.5 Human-in-the-loop machine learning

The assisted-annotation workflow is not merely a pragmatic compromise but an established methodology. Mosqueira-Rey et al. (2023) survey human-in-the-loop machine learning, in which humans and models collaborate so that human input corrects and guides model output. This provides theoretical legitimacy for a system whose suggestions the user reviews, reframing manual annotation from a weakness into a deliberate strategy that improves reliability and yields verified data. Of the three paradigms they distinguish, the system most closely resembles interactive machine learning, in which the user iteratively refines model outputs.

This grounding also shapes evaluation. The appropriate criteria are human-in-the-loop criteria: reduction in manual effort relative to fully manual review, the quality of the resulting annotations, and the number of interactions needed to repair a given class of error. That last criterion is what makes the tracking failures reported in Chapter 5 tolerable rather than fatal, since each is designed to be correctable by a single high-level action rather than frame-by-frame editing.

2.6 Existing systems

Three commercial tools illustrate the practical landscape. Athlete Analyzer offers fencing-specific video review in which the user uploads footage and tags actions by hand, producing hit-zone and pattern statistics from those tags. Its domain alignment confirms demand for structured fencing analysis, but the labelling is entirely manual and the analytics are summaries of what the user entered rather than measurements taken from the video. Dartfish, a mature general-purpose platform, demonstrates the proven value of video analysis across many sports, but is not fencing-specific,

depends heavily on manual operation, and assumes analyst expertise club-level users frequently lack. AI FencingMeter is the closest commercial system to this project's stance, applying automated analysis to smartphone footage, but it reports blade-speed and reaction-time measures for a single fencer's technique rather than the inter-fencer geometry and event structure that this project targets, and its methods are not published, so its claims cannot be assessed.

Together these define the gap this project addresses: fencing-specific tools whose analytics are only as good as what the user typed in, a general platform without fencing specificity, and an automated tool measuring a different quantity by undisclosed means. None computes continuous inter-fencer distance, which is not merely absent from them but impractical to produce by hand at all. That is the substantive argument for automating the measurement layer even when event interpretation stays with the user.

2.7 Synthesis

An apparent contradiction, and its resolution. Vahdani and Tian hold that temporal action detection is unsolved on realistic footage, while Mo reports 89 to 90 per cent accuracy on a harder fencing task than this project attempts. Taken at face value these conflict. They do not, and the reconciliation determines this project's design. Mo's conditions are precisely those the survey identifies as what supervised methods require: roughly 4,000 clips of international competition, captured under fixed cameras with consistent framing. The survey's pessimism is about footage without those properties. Mo therefore does not refute the survey; he satisfies its preconditions, and in doing so demonstrates that the binding constraint is data and capture conditions rather than technique.

A dependency running through three of the four works. Mo's system rests on pose, and Hong et al. establish that pose estimators degrade under motion blur, occlusion and domain shift, which are exactly the conditions elite capture minimises and club capture does not. Mo's result is therefore conditioned on the circumstances that suppress the failure Hong et al. document. The same architecture on single-camera club footage inherits a weaker signal, and the literature offers no evidence about how much weaker. This project's own measurements supply a little: pose succeeded on roughly a third of tracked frames, failing most in the clinches Hong et al. name.

Where that leaves the design. If automation in this domain is conditioned on data and capture this project does not have, the useful question is not how to approach Mo's accuracy but how to be useful without his conditions. Mosqueira-Rey et al. answer it by making human correction a designed component rather than a fallback, and by supplying the evaluation criteria that follow, namely effort saved and interactions needed to repair an error rather than accuracy alone. The four works therefore form an argument rather than a list: the method is available, its performance is conditioned on data quality, the conditions do not hold here, and a human-in-the-loop design is the response.

Two critical reservations. First, the fencing-specific literature is thin, resting substantially on one system, so several arguments here transfer from adjacent sports rather than resting on fencing evidence. Second, and more sharply, following the literature was itself a source of error. Mo's validation of audio as a cue was adopted and the resulting detector failed on held-back footage, because the finding was conditioned on capture quality that the paper had no reason to foreground.

Background work supplies methods together with the circumstances that produced them, and a review reporting the method without the circumstances invites exactly that

mistake. Chapter 5 records the cost, which was one labelling session, and the practice adopted in response, which is that no claim from the literature is carried into the design without being tested on held-back footage of the kind the system will actually meet.

3. Design

Users and domain

The target user is a club-level epee fencer or coach with recorded bouts, no analyst and no budget. Every automated output is a proposal, and the user's time is the scarce resource.

Architecture

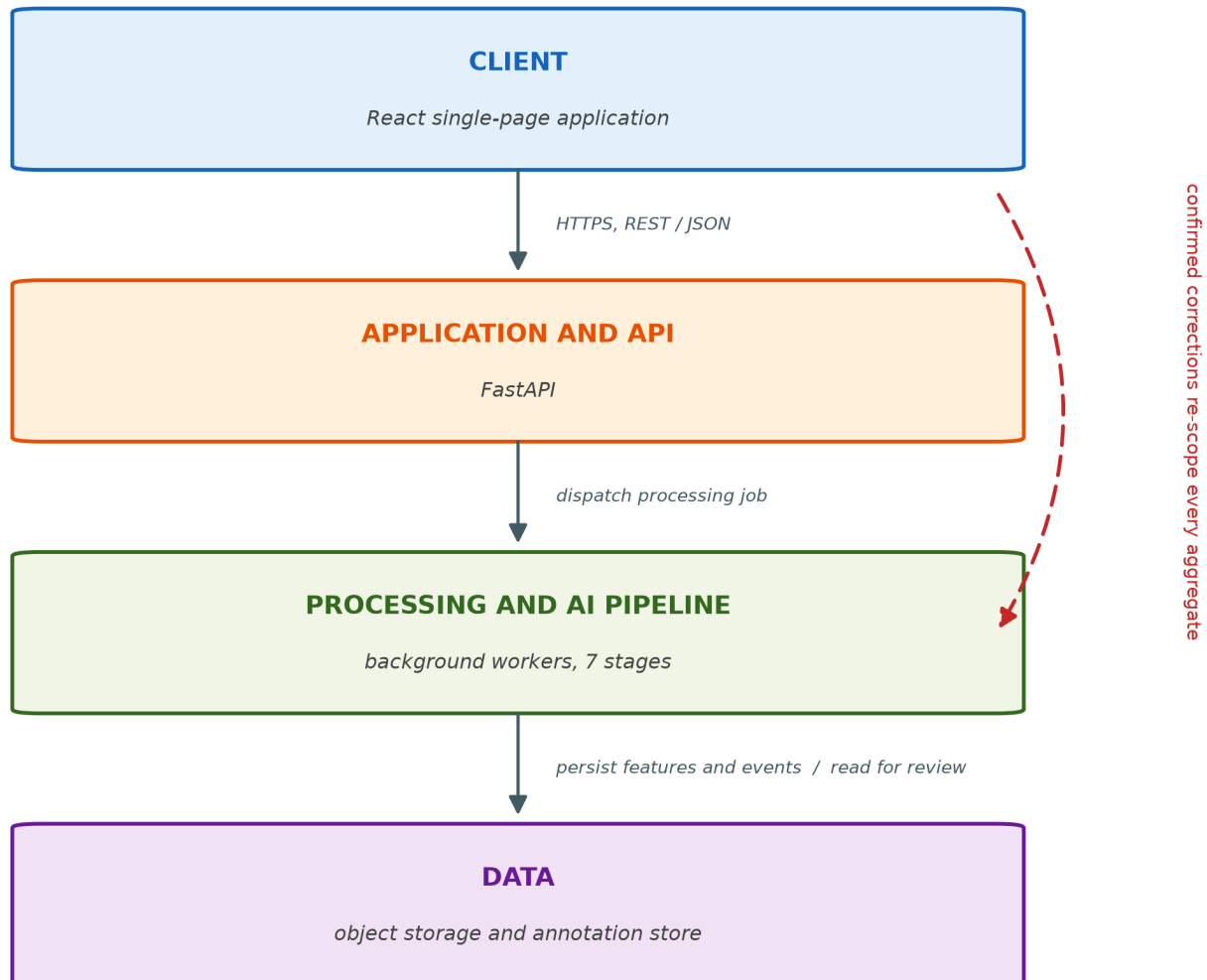


Figure 3.1: System architecture. Four layers and what passes between them. The dashed return path is the only structural element rather than a descriptive one, and it is what makes the workflow human-in-the-loop: confirmed corrections are written back and re-scope every aggregate, so the statistics describe the verified record rather than the system's first guess.

Layer	Contains
Client	upload, piste confirmation, frame-accurate player and timeline, the review loop, results
Application and API	upload handling, job dispatch and status polling, the annotation store, analytics endpoints
Processing pipeline	ingestion; detection and tracking; pose; feature extraction; touch proposal; attribution and profiling; language generation
Data	uploaded video, per-frame CSV, job records, annotations

Two properties matter more than the layering. **Every stage's output is persisted as editable data rather than baked into the rendered video**, which lets the interface read it back. And **the return path is a rescoping rather than a refresh**: confirmed touches bound each exchange, so a correction changes which frames contribute to every downstream aggregate, not merely which events are listed.

Pipeline stages are subprocesses rather than imports. The heavy work is native code, and YOLO, OpenCV and MediaPipe can fail in ways that end the interpreter rather than raise something catchable; in a thread that takes the API and every queued job with it, whereas a subprocess failure is an exit code against one job. The models cost hundreds of megabytes of resident memory, which a subprocess returns on exit. And the pipeline stays a working command line, invoked by the commands that produced the figures here, so no second code path can disagree.

Technology selection

YOLOv8 with ByteTrack for detection and tracking, mature, pre-trained on a generic person class and bundled with a stable tracker; the nano variant runs on a consumer CPU and a larger one substitutes without code changes. **MediaPipe Pose** (`pose_landmarker_lite`), run on each fencer's crop so the model is given an easier problem constrained to the right person. **A hosted large language model** for the written summary, the alternative being a template system that could not adapt as the payload grows.

All three are used pre-trained. No model is trained or fine-tuned: no annotated fencing dataset exists at the scale training requires, and building one is a larger project than this. The contribution is the orchestration, not model development.

The assisted-annotation workflow

Four user actions carry the design, each operating at the level a user already thinks about a bout and repairing many frames of derived data in one interaction: **confirm or correct a proposed touch**, accepting, retiming, attributing or rejecting it; **add a touch the system missed**; **mark a segment tracking-unreliable**, excluding it from aggregates while the raw data is kept; and **re-anchor a slot** by clicking the correct fencer.

The first carries structural weight beyond the touch record. Confirmed touches bound each exchange, so resets and walkbacks are excluded from every aggregate, and those are precisely the periods in

which the referee enters frame. One domain-level act therefore removes most bystander-contaminated data without the user thinking about tracking at all.

Touch detection: why one signal cannot work

The obvious signal is the scoring machine's buzzer, and Mo (2022) validates audio as a fencing cue. Audio alone is insufficient for reasons specific to how fencing is filmed: several bouts run within earshot at a competition and the microphone cannot tell which machine fired; fencers test blades against the floor, which registers identically; and parries produce metallic transients resembling a short buzzer. Each candidate is therefore scored on several features, each included because it discriminates a specific false positive.

Because the design only needs to *propose* candidates, recall matters more than precision, a materially easier target than Mo (2022), whose system decides which fencer scored. Double touches are an attribution problem rather than a detection one: both fencers score within the lockout interval, the event is detected identically, and assignment is already the user's responsibility.

A capability hierarchy, and the annulment argument

Available signals differ by footage type, so the design degrades in tiers rather than failing. Tier 1 is a score change, from a broadcast overlay or venue machine, giving timing and attribution. Tier 2 is a buzzer with geometric corroboration. Tier 3 is geometry alone, for club footage with no visible machine and no usable audio. Reporting per tier is more honest than a single number averaging over categorically different information.

Tier 1 is not merely the most convenient signal but the only one reflecting the referee's decision. A scoring machine registers a valid electrical contact; a score display registers an awarded point. These differ whenever a referee annuls a touch, for corps-a-corps, for covering target, or any other non-valid action.

This justifies the assisted design more strongly than the literature does. That argument rests on models being unreliable, which invites the reply that better models would remove the need for the user. The annulment case is not of that kind. When a referee annuls a touch, the hit occurred, the machine fired, and no point was awarded: **the information distinguishing those outcomes is absent from the recording entirely.** It is a refereeing judgement, not a physical event, and no model can recover it. The detector therefore cannot be correct in principle, only useful.

Piste regions are measured, not drawn

Detections whose feet fall outside the piste are discarded before the identity matcher runs. This matters because the matcher's gates depend on slot history, so on the first frame they accept whatever the detector returns; filtering first means a bystander cannot become a slot's initial anchor and then be defended by the gates thereafter.

The region is derived by sampling frames and clustering where feet actually fall, then shown for confirmation, rather than drawn by hand: a polygon placed by eye raised headline coverage while quadrupling physically impossible distance readings. The user is still asked, because the measurement assumes the two people closest together at the same depth are the fencers.

Evaluation strategy

The template proposes evaluating such a system by accuracy against ground truth and by user feedback. Both are adopted, but neither is sufficient, and the strategy departs from them in three ways the results chapter justifies.

Held-back data rather than accuracy alone. An accuracy figure quoted on the data a system was tuned on measures fit, not capability. Clips are therefore withheld, and the operating point fixed on one applied unchanged to the others. This exposed the first touch detector, which scored respectably on its own clip and collapsed elsewhere; without it the evaluation would have reported a feature that did not work.

Signatures of failure rather than proxies for success. Coverage records whether the tracker committed to a detection, not whether it committed to the right person, so a confident wrong answer raises it. The strategy adds a measure that can only indicate failure: two fencers do not cross on a piste, so a sign change in their relative position is a tracking error, not a fencing event. It needs no threshold and cannot be improved by being wrong more confidently. Evaluation is anchored where possible to a constraint the sport imposes rather than a number the system produces.

Resolution as well as correctness. A metric can be correctly derived and still incapable of distinguishing anything. Each is therefore tested for what it could detect, by block resampling for intervals and by running the same machinery against synthetic series with known biases, so a null is readable rather than merely absent. The template does not suggest this strand, and it is where the methodological contribution sits.

The effort claim is measured by timing an assisted pass against a manual one in the same interface. A user study would measure it better and is further work; this is what can be done without participants.

Three limits are acknowledged in advance rather than discovered: the set is small and single-labelled; there is no frame-level tracking ground truth, so the tracking strand measures commitment rather than correctness; and the effort measurement, on the author alone, is an illustration rather than a population estimate.

Methodology and risk

Development proceeds in two-week iterations, each producing demonstrable output, with commits at green-test states. AI components are de-risked first, so feasibility is established before the interface is built. The principal risks are model performance on real footage, mitigated by the assisted workflow and fallbacks; scope creep, mitigated by separating current scope from further work; and compute cost, mitigated by frame-stride throttling on pose.

Inclusive design

Inclusive design is broader than usability and accessibility. It asks who a system excludes and why, and the answer is rarely that a person lacked some capacity: exclusion is produced by processes that settled on a narrow default through convenience or habit. That radical-inclusion position locates the barrier in the environment rather than the user, and asks that inclusivity be present from the start.

Who this excludes. Para fencers, completely and not incidentally. Wheelchair fencing is fenced from frames fixed to the floor: no footwork, no closing of distance by the feet, no lunge in the sense the pose features model, and every measurement rests on those. Pointed at a Para bout the system would not fail visibly, which is worse: a flat distance series, no proposed touches, a profile at parity. **Confident nonsense on a population it was never designed for is a worse exclusion than a refusal.**

What was done about it. The system now refuses: if neither fencer's position varies by more than 0.25 m, the per-fencer axes are withheld and the reason given. The threshold comes from the four clips, whose smallest spread is 0.66 m, and is calibrated from one side only, no wheelchair footage being available. This does not make the system inclusive; it makes it honest about the boundary of what it measures.

A population assumption is baked into every distance figure. The pixel-to-metre scale assumes a fencer height of 1.75 m, a men's senior average, applied to every bout. On a women's or junior bout every distance is wrong by the ratio of true height to assumed, invisibly, because it scales everything consistently. A default chosen for convenience became a structural exclusion, unnoticed until the project was examined this way.

Data provenance and bias. The models were selected for availability, itself a bias. YOLOv8's person class comes from COCO, a convenience-collected web dataset the project neither audited nor controls; the documented failure of facial-analysis systems trained on predominantly pale-skinned cohorts is the standing example, and nothing here establishes the detection stage is free of an analogous skew. The evaluation set compounds it: four clips, three elite, all labelled by the author.

What the design gets right, though neither was motivated by inclusion. The review loop is keyboard-driven, every decision one key, removing the precise pointing a scrubbing interface demands. And colour is never the only carrier of meaning: the fencers are blue and amber, which survives the common red-green confusions, but each is also labelled and positioned consistently, and every state shown as a colour is also a word. Both were arrived at for legibility, which is the point radical inclusion makes about process: getting somewhere by accident is not a method.

What was not done. No disabled user was consulted, no assistive technology tested, no stakeholder outside the author's own club involved. The participatory element is absent. The stated user, a club fencer with no budget, is an economic widening of access rather than an inclusive one, and the two should not be conflated.

Ethics

Evaluation footage is publicly available video used solely for non-commercial academic evaluation, not redistributed, with rights retained by the uploaders. It shows identifiable individuals and the system extracts position and pose data about them; no attempt is made to identify anyone by name, no biometric identity model is used, and the system tracks anonymous slots. If the full system stored profiles tied to named individuals, informed consent and a retention policy would be required under UK data-protection norms. Any user testing will be carried out with consenting healthy adults.

4. Implementation

This chapter records what was built and where the direction changed, because on this project the changes of direction carry most of the evidence.

Pipeline

Per frame: YOLOv8 runs with ByteTrack filtered to the person class; detections are filtered by the piste region; a custom FencerTracker assigns them to two persistent slots; MediaPipe runs on each slot's crop; distance is measured front-foot to front-foot where pose is available and bottom-of-box otherwise, normalised to metres by mean bounding-box height against an assumed 1.75 m fencer. Pose runs every third frame, trading sample frequency for a roughly threefold reduction in wall-clock time.

Front foot to front foot rather than centre to centre because arm and weapon extension distort a bounding box in a way unrelated to where the body is, and because the gap between front feet is what coaches mean by distance. The metre scale is approximate and reported as an estimate, consistent enough for comparison within a bout.

Identity tracking, in four iterations

The first version locked onto the two highest-confidence ByteTrack IDs and accepted only those thereafter: robust against bystanders, extremely fragile to ID reassignment, losing both fencers for over 95 per cent of frames.

The second replaced strict IDs with **spatial continuity**, assignments minimising total movement from each slot's last known position. Coverage recovered, but the matcher accepted any bystander who happened to be the second most confident detection.

The third added two **gates**, rejecting candidates too far from a slot's last centre or too different in height. Set strictly these dropped coverage to about 55 per cent by rejecting legitimate motion; relaxed, they gave about 82 per cent while filtering obvious bystanders, and cut maximum observed inter-fencer distance from 11.27 m to 6.60 m, confirming they excluded the wrong-target outliers inflating the estimate.

The fourth added the piste filter and a **constant-velocity motion model**. The motivating case is a referee walking into the position a fencer occupied a moment ago: under position-only gating that is an excellent match, since the gate asks only whether the candidate is near where the slot last was; under prediction-based gating it is poor, because the fencer was moving and the referee is not where that motion leads. The predicted position anchors both the gate and the assignment cost, so one change improves rejection and allocation together. Two guards proved necessary after the first version failed on footage: velocity is estimated only from commits within three frames of each other, since differencing positions further apart measures displacement over the gap; and a slot unmatched for thirty frames discards its history, because one whose stale history rejects every candidate can otherwise never recover.

Chapter 5 reports a fifth change, to candidate selection, which turned out to be the cause of both documented failure modes.

Attribution and lunge proposals

`detect_scorer.py` reads the scoring machine's lamps. Lamps rather than a score overlay, because broadcasts carry both but club footage carries only the machine. Whole-frame saturated-colour counts rather than a region around the machine, because club footage is hand-held and follows the action, so the machine drifts out of frame and a fixed region is empty much of the time. A local baseline two to three seconds earlier removes what is permanently red in shot: an exit sign, sponsor banners, the piste. It is never permitted to propose touches, only to attribute ones already found, since the lamps fire whenever the circuit closes, including when fencers test weapons just after a touch. Taking touch times as given removes that failure class rather than filtering it. This is classical colour thresholding and adds no fourth model.

`detect_lunges.py` proposes lunges from the ratio of ankle separation to hip height, calibrated on the first five the user confirms on the bout being watched. It does not transfer a threshold between bouts, because Chapter 5 shows transfer is what fails. Calibrating from confirmed labels reuses the correction mechanism the design already depends on rather than adding a demand on the user.

The language model, and the constraints placed on it

The third pre-trained model turns the collected statistics into a written summary.

`generate_summary.py` is a separate command-line stage, decoupled from the video pipeline: it reads the per-frame CSV, aggregates it into bout-level statistics, embeds that payload as JSON in a structured prompt, and calls a hosted large-language-model API. Output is constrained to a fixed structure of a summary paragraph, observed-tendency bullets each tied to a concrete number, suggestions to explore, and a data-caveats paragraph.

Two design points matter more than the integration. The prompt carries explicit **honesty constraints**: the model is told that distances are estimates, that aggregates are scoped by confirmed touches, that cumulative movement totals are withdrawn, and that it must never invent touches, scores or events absent from the data. This addresses the known risk of fabricated plausible specifics. And generation is **never automatic and is cached**: a hash of the model identity and both prompts is stored alongside the output and the call skipped when nothing has changed, so re-running does not re-bill.

The summaries are constrained by what the pipeline can honestly supply rather than by the model. They grow as the payload does, since the architecture is unchanged as confirmed touches, in-play scoping, tempo and the per-fencer profile arrive. Seventeen unit tests cover the statistics computation, prompt construction and cache behaviour with the API call mocked, so the suite needs neither a network connection nor a key.

The application layer

Every job is a JSON file on disk, so a restart loses nothing: jobs caught mid-run are marked interrupted rather than left claiming to run, and are not retried automatically, since a video that crashes the pipeline would otherwise crash it on every start. Progress is parsed from a machine-readable line the pipeline prints rather than through a callback, which would have required importing the pipeline and undone the subprocess separation.

Two interface decisions are worth recording. Summary generation costs a paid API call, so it is never automatic; what the application layer changed is that the user presses a button rather than being handed a command to type. And re-anchoring, the one action that changes tracking rather than interpretation, can only take effect on a reprocess, so the interface queues that reprocess and writes the result to a **new** bout, leaving the original intact for comparison.

The review interface

The interface is a working instrument, and three decisions in it were reversals of what the first version did.

Colour carries meaning and the chrome does not use it. The original palette took the Fencer 1 blue as its general accent, so the primary buttons, the progress bars and one of the two fencers were the same hue. Since the two fencers are identified by colour throughout, that made the most information-carrying signal in the interface indistinguishable from decoration.

The screen stopped presenting everything at once. Six bordered panels of equal visual weight meant nothing was more important than anything else, and the single act the interface exists for competed with five things the user was not doing. The working column now holds only the video, the timeline, the decision loop and the record it produces; everything else moved into a tabbed column grouped by the question it answers, with the corrective tools separated out because they are used rarely and never during a review pass. Explanatory notes are folded behind single-line disclosures rather than deleted: nearly every caveat here was paid for by a measurement that went wrong, but printed simultaneously they produced a screen that was mostly prose, and a caveat inside a wall of text is read no more carefully than an absent one.

A **review queue** holds the playhead at each proposal in turn and takes one keystroke as the answer. This is less a new capability than the removal of a cost the earlier interface imposed by accident: a table is the right shape for checking work already done and the wrong shape for doing it, since finding the next undecided row and waiting for the seek repeats once per proposal. That is precisely the cost the design claims assisted annotation removes, so leaving it in place would have understated the system in its own evaluation.

Testing

The system is supported by 673 automated tests: 367 over the pipeline, 184 over the application layer and 122 over the interface, plus two end-to-end tests driving the real pipeline on a synthetic video. The end-to-end pair exists for what unit tests structurally cannot reach: the stages are joined by filename conventions rather than return values, and a bout identifier is assembled in one module and taken apart in another. It found two defects on its first run.

Tests are written against pure data behaviours where possible, so the suite runs in seconds and adds no friction; regressions get a test before the fix, and several were confirmed by reintroducing the defect afterwards. The limits of this are recorded honestly in Chapter 5: every defect that mattered on this project passed its tests.

5. Evaluation

Four independently sourced epee bouts, three broadcast and one club recording, each about three minutes. Twenty-seven touches were hand-labelled by the author, a competitive epee fencer, with the scoring fencer recorded. Clips were held back from tuning.

The audio detector failed to generalise, and geometry replaced it

The first touch detector scored candidates from a band-passed audio envelope with geometric corroboration. Tuned on clip 3 it reached F1 0.79; applied without retuning to a held-back clip it collapsed:

	clip 3 (tuned on)	clip 2 (held back)
Precision	0.79	0.21
Recall	0.79	1.00
F1	0.79	0.35
Corrections vs manual	6 vs 14	15 vs 4

Fifteen corrections against a manual baseline of four means the detector produced nearly four times more work than labelling by hand, and confidence lost all ranking power: real touches scored 0.74 to 0.79 while false positives reached 0.91.

Three diagnostics established this was not a badly chosen constant. The noise floor differs by an order of magnitude between a quiet club hall and a broadcast, and five threshold rules each either flood the noisier clip or find nothing in it. A percentile threshold flags a fixed fraction of frames rather than a number of events, so candidate count followed recording length. And searching every frequency band gave a best separation ratio below 1.0 on both clips: the weakest touch is quieter than ordinary background even in the optimal band. On clip 3 the loudest tonal component at each touch sat at a different frequency every time. A scoring machine has one pitch; that scatter indicates the buzzer is not reliably in the recording, and that the detector had been finding blade contact which happens to correlate with touches.

Geometry alone, using features the pipeline already produced, scored F1 0.86 on both clips with identical settings. The audio was not merely unhelpful but harmful, generating candidates the geometry had to filter. The geometric signature follows from how fencing works rather than from a fit: a touch requires closing to scoring distance, and the referee's halt sends both fencers back to their guard lines, so they separate. Distance is also robust to panning, because a pan shifts both fencers together and distance is a difference between them. Candidate rate now tracks content rather than recording length. Across the three 720p clips the detector reaches F1 0.85.

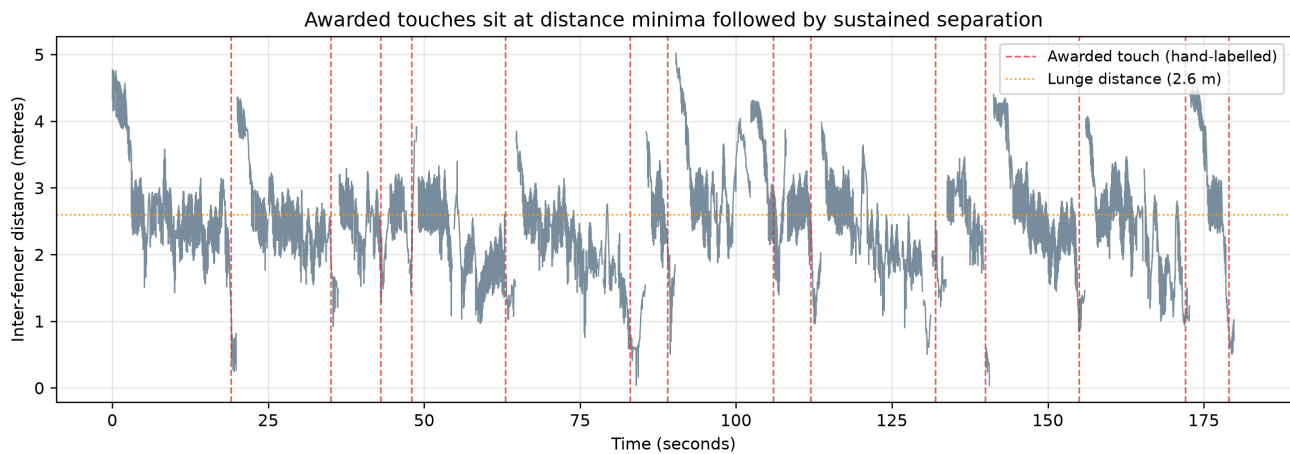


Figure 5.1: Inter-fencer distance across clip 3, with the fourteen hand-labelled touches marked. The touches sit at local minima followed by sustained separation, which is the pattern the detector looks for.

Figure 5.1 evidences the claim that the geometric signature follows from the sport rather than from a fit, and shows its limits. Every labelled touch coincides with a minimum, but not every minimum is a touch: the fencers close to scoring distance repeatedly without a hit registering, which is why the detector is tuned for recall. The last touch, at 179 s of a 180 s recording, is the clearest failure in the plot: its separation window runs past the end of the footage, so the strongest feature cannot fire. An edge effect rather than a detection failure.

Three points follow. **A single-clip evaluation establishes nothing:** the audio F1 of 0.79 looked like a working feature and was an artefact of the clip it was tuned on. **The more sophisticated approach was the worse one:** audio followed the literature, required band calibration and an external dependency, and was beaten by a local minimum in a signal already being computed. And a negative result was necessary to reach the positive one, since geometry alone was not tried until audio had been shown to fail.

The candidate cut: one cause behind both tracking failure modes

Two failure modes were treated for most of this project as inherent limits of single-camera tracking: wrong-target capture by background people, and close-range identity flicker. Investigating why a correctly aimed correction repaired nothing showed they shared one cause, upstream of both the correction and the gates.

FencerTracker.select narrowed each frame to the two most **confident** detections before any anchor, gate or correction was consulted, on the assumption that the two fencers are the two people the detector is most sure about. On competition footage that is measurably false, because the referee stands on the strip and survives the piste filter by construction. Sampling clip 2's six-second bystander capture every twentieth frame: three detections lie inside the piste region in every sampled frame, and the fencer a user would click is detected in every one, yet the confidence cut discards that fencer in 9 of 16. Where the capture becomes visible the three confidences are 0.899, 0.896 and 0.886. The tracker was arbitrating between three people on margins that are noise.

This also explains why re-anchoring could not repair the failure, which had been read as a limitation of the action: the cause is sustained over six seconds and the correction momentary, so the slot is handed the correct fencer and has lost them by the next frame.

Selecting the two candidates nearest each slot's predicted position instead, which costs nothing since the predictions already exist for the gates:

Clip	Coverage before / after	Mix-ups before / after	Touch F1 before / after
1	92.9% / 93.5%	2 / 0	0.80 / 0.80
2	98.0% / 98.0%	15 / 0	0.86 / 0.86
3	97.4% / 97.4%	0 / 0	0.86 / 0.86
4	73.7% / 79.8%	114 / 54	0.60 / 0.60

Mix-ups are single-frame position jumps exceeding 1.5 m, a direct signature rather than a proxy: no fencer crosses metres of piste between consecutive frames.

An earlier version reported clip 4's touch F1 rising from 0.67 to 0.77. That figure was read at clip 4's own best threshold rather than the operating point tuned on clip 3, which is the protocol applied elsewhere. Choosing a threshold by looking at the held-back clip is the error the audio detector was diagnosed with, and quoting a benefit obtained that way would repeat it in the project's favour. At the stated point clip 4 scores 0.60 either way.

What this costs the argument. Wrong-target capture was presented as evidence for the assisted design: vision fails this way, therefore the user must repair it. On three of four clips it now does not occur. It was a defect in candidate selection, not a limit of single-camera tracking, and the honest reading is that the project spent a long time designing around a bug. The argument survives on the failures that remain.

Slot identity, and a check that needed no threshold

Clip 4's Fencer 2 reported +7.08 m of net displacement, which is impossible: play resets to the guard lines after every touch. Slot identity had not held. Two fencers do not cross on a piste, so a sign change in the difference between tracked positions is not a fencing event but the tracker exchanging which fencer each slot follows. Counting those separates the set completely: clips 1 to 3 record zero swaps, clip 4 swaps 14 times and holds Fencer 1 on the left in 26.9 per cent of frames. Clip 4's fencers come within 0.09 m against 0.26 to 0.65 m elsewhere, which is where the matcher has nothing left to separate them by.

Two experiments previously recorded as “worse”, cropping to the piste and masking the gallery, swap 57 and 64 times against a baseline of 14. The masked variant reports net displacements passing the implausibility test comfortably while its slots exchange fencers 64 times. **A believable-looking number is not evidence that identity held.** The system now warns on any swap, needing no threshold, because fencers do not cross.

Attributing a touch to a fencer

Evaluated leave-one-out against the scorer column already in the ground truth, so it required no new labelling:

clip	three-way (left/right/double)	green lamp alone
1	3/3	3/3
2	2/4	4/4
3	9/14	14/14
4	4/6	6/6
all	18/27 (67%)	27/27 (100%)

The green lamp is reliable and the red is not, and nearly every three-way error is a red-lamp error, red being contaminated by everything permanently red in frame. The claim is therefore narrower than “the system says who scored”: **it always identifies whether one named fencer was involved**, which fully determines the 11 touches of 27 where they were not and reduces the other 16 to a two-way decision. The colour-to-side mapping is confirmed once per bout by the user, since nothing in the image indicates it.

Lunge detection works per bout and does not transfer

An operating point fitted on clip 3 reaches F1 0.22 on clip 2 while firing on a quarter of all windows. Stated directly: clip 3 calibrates to a stance ratio of 1.902 and clip 2 to 2.706, a 42 per cent difference in what counts as a lunge-like posture. Calibrating on the first five confirmed lunges and testing only on those that follow, clip 3 reaches F1 0.75 on 25 held-out lunges against that 0.22 baseline.

Only that row carries weight; clip 2’s test sets contain two and four lunges. Precision is a lower bound, since a firing on a real but unlabelled lunge counts against it, and roughly 40 wide-stance episodes per minute were measured against about 10 labelled ones. The claim is not that lunge detection works, but that per-bout calibration works where transfer does not.

What closing share can resolve, which is nothing

Closing share, the proportion of moving frames spent reducing distance, survived scrutiny on the reasoning that counting direction is robust where summing magnitudes is not. That reasoning is sound and says nothing about resolution. All eight fencer-clip values fall between 48.3 and 52.8 per cent.

Frames are strongly autocorrelated, so treating each as independent would make every figure look significant. Resampling contiguous blocks of a third of a second instead, **not one of the eight is distinguishable from a coin flip**. A null is only readable alongside what the method could have detected, so the same machinery was run against series with known biases: on three minutes it detects a 60/40 tendency in every trial, 55/45 in 75 per cent and 52/48 in 30 per cent, against a false-positive rate of 11 per cent. Every measured value sits where the method finds a real tendency about a third of the time, so the measurements are consistent both with these fencers being balanced and with tendencies too small for three minutes to resolve.

This is the fifth measurement in the project to survive its own reasoning and fail a test of what it can resolve, and the pattern is now specific enough to state as a rule: **a metric’s derivation establishes what it means, never what it can detect, and the second question needs its own measurement.**

Characterising a fencer without accumulating error

Cumulative push and pull totals were withdrawn because they accumulated (Appendix D): a fencer re-acquired at a new position after a dropout contributes a one-sided step that never cancels, so clip 3's Fencer 2 accumulated +23.01 m against an endpoint difference of -0.94 m, and no clamp repairs it because the truncated steps reach 5.4 m in a single frame. What failed was the operation, not the series. An instantaneous reading averaged over thousands of frames absorbs a dropout; a running sum banks it permanently.

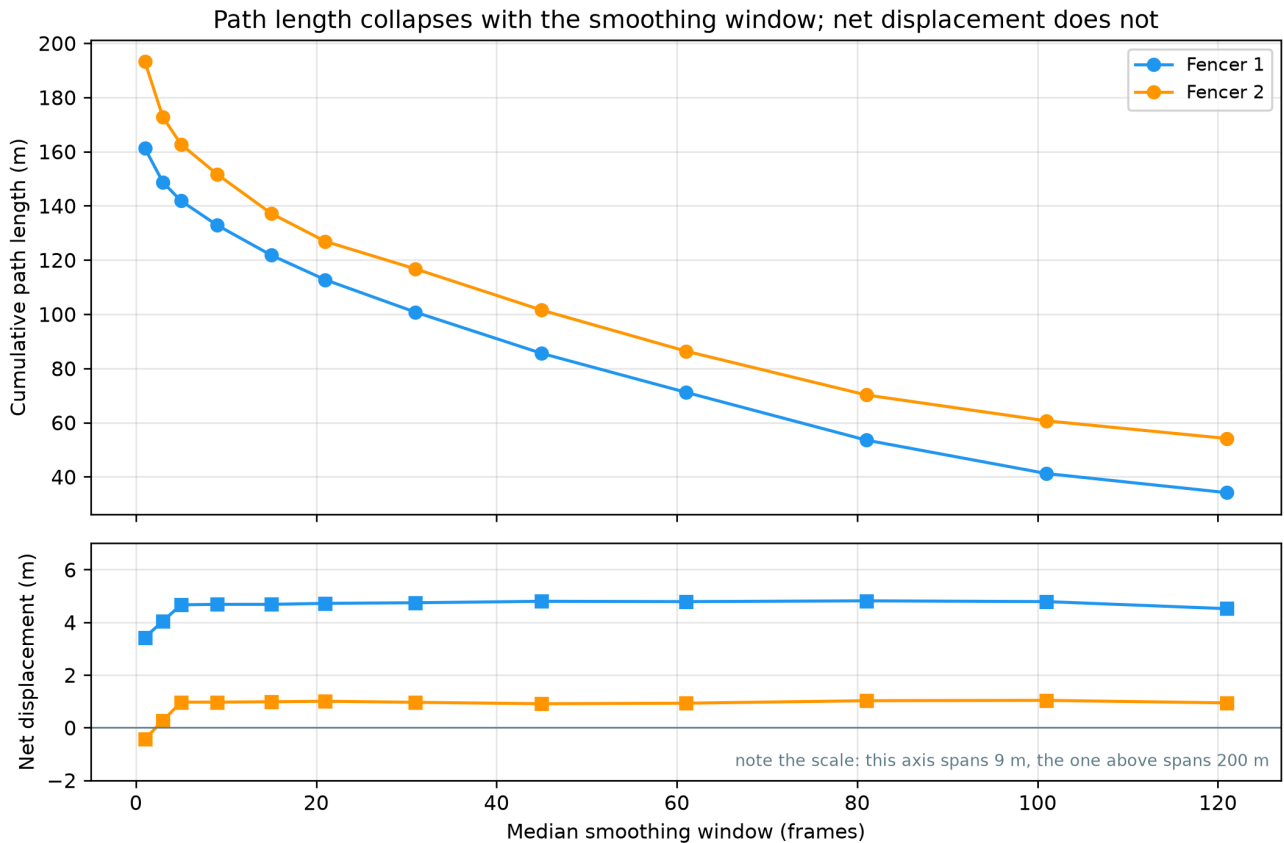


Figure 5.2: Path length and net displacement across median smoothing windows, clip 3. Both are computed from the same smoothed series at each window.

Figure 5.2 is why the totals were withdrawn rather than qualified. Path length falls monotonically as the window widens with no asymptote, so any figure quoted is a statement about the smoother rather than the fencer. Net displacement, from the same series, moves once and holds. The contrast within one plot makes the diagnosis certain, since both see identical input and only one is stable under it.

A six-axis per-fencer profile was therefore built in which no axis accumulates: three are a mean or quantile of position, three are counts over confirmed touches. On clip 3 the two fencers separate on a reading no earlier number expressed: Fencer 2 occupies nearly twice the ground while Fencer 1 scores at the longer distance. Whether that describes the bout is an outstanding verification rather than a finding.

The profile introduces the first outright refusal in the system. All six axes assume slot identity held, and on clip 4 it did not; a profile there would describe the tracker while looking as convincing as one that did not. Displaying a figure alongside its caveat is sound for a number and not for a shape: a radar with a footnote is still read as a radar.

Measuring the effort the design claims to save

The central claim is that confirming proposals costs less effort than labelling from scratch, and nothing measured it. A review queue and a manual mode were built as the two conditions, the manual mode withholding proposals entirely rather than dimming them, since a visible suggestion has been read by the time the user decides to ignore it. Both time themselves. Skipped items are excluded from the denominator, since counting them would improve the rate in proportion to unanswered questions; and the rate is per decision, because assisted review answers one question per proposal while manual logging creates one entry per touch found.

No such pass has been recorded, so this reports a method and not a result. The instrument exists and has been exercised end to end; the figure requires a person who can tell a lunge from a wide stance to sit down with a bout.

Evaluating the models as components

The evaluation above measures what the pipeline produces. Examining each model separately exposes what those figures hide.

Detection is measured by commitment, not correctness. Coverage records whether the tracker committed, so a confidently wrong detection raises it. There is no frame-level detection ground truth, which is why the side-swap count exists.

Pose is reported as a configuration, not a capability. The figure quoted, roughly a third of tracked frames, is governed mainly by the frame stride, and says little about how often pose succeeds when attempted.

The language model was not evaluated at all, the weakest point in the original strategy. Its documented failure is fabrication: fluent prose containing figures never in the input, the fluency being what stops a reader noticing. `audit_summary.py` checks every number in a summary against the payload given. Across two cached summaries: **58 figures, none unsupported**, the one flag being the parity reference “50 per cent” rather than a data claim.

That is only readable alongside what the audit could catch, so it was run against fabricated figures.

The first version detected none. It treated a difference or sum of any two payload values as legitimate derivation, which seemed conservative and let sixty values generate combinations covering the small numbers densely. Restricting derivation to direct matches raised detection to 90 to 98 per cent for any figure with a decimal or above 20, at the cost of occasionally flagging a legitimate one. Small integers stay weakly covered, a payload this size containing most of them.

The honest result is narrower than “the summaries are faithful”: **no fabricated decimal figure appears in either**, on an instrument measured to catch such figures nine times in ten. This is the fifth occasion on which a method’s derivation established what it meant and a separate measurement was needed to establish what it could detect.

Coverage and remaining failures

Coverage, meaning both fencers detected and assigned to stable slots, is 93 to 98 per cent on three clips and 79.8 per cent on clip 4. The remainder is mostly a fencer partly off-screen, a gate rejecting a referee, or flicker dropping a slot. None introduce wrong data; they reduce the contributing samples.

6. Conclusion

Against the original aims

The project set out to orchestrate several pre-trained models into a system that helps a fencer analyse a bout, on the argument that unreliable computer vision becomes useful when a person can repair each class of error cheaply. Three models were orchestrated in a genuine chain rather than demonstrated side by side: detections bound the region pose works in, pose landmarks supply the ground reference the distance series is measured from, that series drives touch proposal, confirmed touches scope every aggregate, and those aggregates are what the language model interprets. Each stage consumes the previous one, and a defect in an early stage propagates, which Chapter 5 demonstrates rather than assumes.

The measurable outcomes are strongest where the evidence is strongest. Touch detection reaches F1 0.85 across the three 720p clips against hand-labelled ground truth, and confirming its proposals costs six corrections where labelling from scratch would cost twenty-one. Tracking holds both fencers in 93 to 98 per cent of frames on three of four clips. Attribution, which the detector could never supply, identifies whether one named fencer was involved in all twenty-seven labelled touches. These are the claims the project can defend.

The central claim is not among them. That assisted review is faster than manual review has never been measured. The corrections figures standing in for it weight a rejection and a manual addition equally, which is known to be false: rejecting a proposal is one keystroke on an event already located, while adding a missed touch requires scrubbing to find it, which is exactly the labour the system exists to remove. The instrument that would settle it now exists and has not yet been used. This remains the largest gap between what the design argues and what the evidence supports.

What the project learned about its own method

The most transferable outcome is methodological, and it recurred often enough to be a finding rather than a series of accidents. **Five separate defects passed their tests while measuring the wrong thing.** A distance-band classifier used thresholds invented rather than derived, and inverted a clip's headline reading once corrected. A movement metric accumulated its own smoothing and reported twenty-four metres of travel on a fourteen-metre piste. A resolution ablation varied a quantity the detector never sees, so its flat result was guaranteed in advance. A piste polygon placed by eye raised coverage while quadrupling the count of physically impossible readings. And the tracker chose between three people using confidence margins of about 0.01, which is noise, for the whole life of the project.

Each was invisible to unit testing because each was implemented exactly as specified. What exposed them was measurement against reality: a physical constraint the sport imposes, a held-back clip, a metric chosen because it could only move one way. The rules the project ended with are that a parameter should be derived from data rather than chosen; that a single-clip result establishes nothing until a held-back clip agrees; and that a headline metric improving is not evidence when a second metric can move the opposite way. Clip 4 demonstrated that last point twice, with coverage rising while wrong-target capture rose too.

The corollary is uncomfortable and worth stating. Wrong-target capture was presented for most of the project as evidence for the assisted design. It was a defect in candidate selection, and on three of four clips it no longer occurs. The design argument survives on the failures that remain, but a substantial amount of design effort went into accommodating a bug.

The same pattern held for presentation. The interface was rebuilt four times, and not one of the intermediate failures showed up in a test, because what the tests assert is that the right information is reachable and in every version it was.

Limitations

Four qualify everything above. There is **no user evaluation**, so the effort claim is unevidenced.

Pose does not fully earn its place: it improves the distance measurement, but its headline application to lunge detection produced an operating point that did not survive a change of camera, and only per-bout calibration made it usable. **The sample is four clips and one labeller**, three of them competition footage, so figures are plausibly optimistic for the club setting that is the target use case. And **there is no frame-level tracking ground truth**, so coverage measures whether the tracker committed to a detection, not whether it committed to the right person.

Future work

The immediate work is evaluative rather than technical. A user study measuring review time against a manual baseline would settle the central claim, and weighting corrections by their true cost would repair a metric this report currently qualifies. Both need participants rather than code.

Technically, three directions follow from measured failures rather than from ambition. An appearance-based re-identification embedding is the remaining lever against wrong-target capture, since a spatial filter cannot separate a referee standing on the piste from a fencer. Clip 4's Fencer 2 still reports an implausible seven metres of net displacement, which the candidate-selection fix did not explain and which therefore has a cause still unidentified. And lunge detection rests on two clips and one labeller, so more labelled footage would establish whether per-bout calibration generalises or merely fitted twice. Beyond that, reading the score overlay with OCR would add a fourth pre-trained model and recover the single-versus-double distinction the lamps cannot reliably provide.

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Appendices

Appendices are excluded from the word count and carry the evidence supporting Chapter 5.

`final/final_report.md` holds the full derivations; the appendices below are the subset a reader needs in order to check the claims made here.

A. Source code overview

Three directories. `code/prototype/` is the pipeline, every stage a standalone command-line program that the web layer invokes rather than reimplements, so no second code path can disagree with the figures in Chapter 5: `run_detection.py` (detection, tracking, pose, distance and the annotated render), `derive_piste.py` (measures a piste region from footage), `detect_touches.py`, `detect_scorer.py`, `detect_lunges.py`, `fencer_profile.py`, `in_play.py` (scopes metrics to playing time), `tempo.py`, `generate_summary.py`, and the `evaluate_*.py` scorers against ground truth. `code/backend/` is the FastAPI application: endpoints, the background job runner, the annotation store and disk accounting. `code/frontend/` is the React interface.

B. CSV schema

The per-frame CSV has columns `frame`, `time_s`, `distance_raw_m`, `distance_smooth_m`, `method` (pose or bbox), `f1_advance_m`, `f1_retreat_m`, `f2_advance_m`, `f2_retreat_m`, `f1_pos_m`, `f2_pos_m`, `f1_stance_m`, `f2_stance_m`, `f1_hip_height_m`, `f2_hip_height_m`.

The four advance and retreat columns are **retained but not reported**. They are the withdrawn cumulative totals, kept because Appendix D's analysis of why they fail is reproduced from them, and because removing a column from a schema earlier results were written against would make those results unreadable. The position columns are what the reliable per-fencer figures derive from; the stance and hip-height columns are what lunge proposals calibrate on.

C. Footage attribution

To be completed before submission with per-clip source URLs, uploaders and access dates. All four clips are publicly available video used solely for non-commercial academic evaluation and are not redistributed with this report.

D. Measurement investigations

Held in full in `final/final_report.md`, Appendix E. The material Chapter 5 depends on:

Why the cumulative movement totals were withdrawn. Replaying the recorded position series through the accumulator's own logic accounts for the divergence exactly. Clip 3's Fencer 1 shows an endpoint difference of +4.71 m against +13.96 m accumulated; Fencer 2 shows -0.94 m against +23.01 m. Residue left unbanked is 0.015 m, so nothing else contributes. The per-frame clamp fires on 63 and 65 frames of 5,246, and those frames are nearly balanced in count, 30 retreat-side against 35 advance-side, yet net to -23.94 m, because the 35 per cent of them falling within a smoothing window of a lost-tracking frame carry -25.14 m while the remainder roughly cancel. A symmetric out-and-back glitch crosses the clamp twice in opposite directions and cancels; a fencer re-acquired at a new position after a dropout is a one-sided step that does not. No clamp threshold

repairs this, because the truncated steps reach 5.4 m in a single frame. The unit test asserting the endpoint identity could not have caught it: its own comment records that its steps were chosen to stay inside the clamp threshold.

Camera motion was tested and refuted as the cause. Pan displaces both fencers alike, while “toward the opponent” points in opposite directions for them, so pan makes their net figures move oppositely. Clip 3’s were both positive. Compensation by Lucas-Kanade optical flow with the tracked boxes masked out was implemented and validated by recovering a synthetic pan exactly, and reduced a synthetic pan bias of +8.33 and -8.30 m to +0.03 and +0.03 m. On real footage it did not repair the totals, which is what identified accumulation rather than panning as the cause.

The resolution ablation was invalid by construction. It varied output encoding rather than the resolution the models see, so a flat result was guaranteed in advance and established nothing about robustness to input quality.

E. Development log highlights

Held in `final/final_report.md`, Appendix D, and in the repository’s commit history, which is the primary record. `TODO.md` preserves completed items rather than deleting them, so the audit trail of what the project attempted is intact.